

Optimizing Paper Submission Recommendation System: A Cross-Domain Approach using Transformer Models

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Project Overview

In response to the challenges faced by researchers in selecting appropriate journals for their submissions, this study investigates advanced methodologies to enhance journal recommendation systems.

The baseline approach for this research builds upon the SimCPSR model, which utilizes a fine-tuned Distil-RoBERTa transformer model for paper submission recommendation tasks. The methodology involves a two-stage process: fine-tuning the language model with a contrastive learning technique and subsequently employing it for downstream classification tasks. The fine-tuning step optimizes semantic embeddings by leveraging paper metadata, such as titles, abstracts, and keywords, paired with journal-specific attributes like aims and scopes, using a contrastive learning framework. This approach not only improves the alignment of paper representations with their respective journals but also enhances the model's ability to distinguish between similar yet distinct topics, leading to more accurate and context-aware recommendations. On the next stage, the approach incorporates two classification models. The first relies solely on metadata from the paper, including its title, abstract, and keywords. These features are combined in various configurations and encoded into high-dimensional embeddings using the transformer model, which then predicts the most relevant journals. Meanwhile, the second model builds on this by incorporating journal-specific information, such as “aims and scopes,” alongside the paper metadata. By merging embeddings from both sources, this model aims to capture richer semantic relationships, potentially enhancing recommendation accuracy. The framework demonstrated state-of-the-art results in domain-specific applications, particularly within the Computer Science field.

Building on this baseline, significant advancements are introduced to improve performance and adaptability. These include leveraging advanced transformer models such as all-MiniLM-L6-v2 and BAAI/bge-small-en-v1.5, which are optimized for generating high-quality embeddings. The study also expands the scope to evaluate cross-domain performance by incorporating a new Biology dataset alongside the Computer Science dataset. Two distinct strategies are assessed: direct use of pre-trained embeddings versus domain-specific fine-tuning. The primary objective of this study is to evaluate whether these proposed improvements can address limitations of the baseline, such as overfitting and limited generalization, while enhancing the recommendation system's accuracy and robustness across domains. This framework provides a comprehensive foundation for optimizing journal recommendation systems and advancing their adaptability to diverse academic fields.

In the end, the study demonstrates that advanced transformer-based models significantly enhance journal recommendation accuracy and cross-domain adaptability. Pre-trained

embeddings, particularly from BAAI/bge-small-en-v1.5, outperform fine-tuned models while reducing computational requirements. Surprisingly, incorporating journal-specific metadata, such as aims and scopes, yielded minimal improvements, with core paper metadata proving sufficient for effective recommendations. However, overfitting to domain-specific features and challenges in cross-domain generalization highlight areas for improvement. Future work will focus on domain adaptation techniques, such as adversarial training, and refined fine-tuning strategies to improve model robustness and adaptability across diverse academic fields.

Abstract

The exponential growth in academic publications across various domains has created challenges for researchers in identifying suitable journals for their work. This study presents an advanced paper submission recommendation system leveraging state-of-the-art transformer-based language models, including all-MiniLM-L6-v2 and BAAI/bge-small-en-v1.5. Key objectives include evaluating the performance of these advanced models against the baseline Distil-RoBERTa, analyzing the impact of domain-specific fine-tuning on classification performance within and across domains using Computer Science and Biology datasets, and comparing the effectiveness of fine-tuned models versus directly leveraging pre-trained embeddings for classification tasks in isolated datasets. Our findings highlight that pre-trained embeddings from advanced models provide robust cross-domain adaptability, outperforming fine-tuned models in certain scenarios. Contrary to expectations, incorporating journal-specific metadata, such as “aims and scopes,” offered narrow improvements. Despite these advances, overfitting to domain-specific features limits generalization. Future work will explore domain adaptation techniques, such as adversarial training, and refined fine-tuning strategies to enhance cross-domain applicability.

Keywords: Recommendation System, Large Language Model, Natural Language Processing.

Introduction

The rapid growth of scientific research has led to a significant increase in the number of academic papers published across various fields. This surge presents a challenge for researchers, particularly in selecting appropriate journals for paper submission. A well-chosen journal not only enhances the visibility of the research but also increases the likelihood of acceptance. However, identifying the most suitable journal, especially in specialized domains with numerous publication venues, remains a non-trivial task.

Paper submission recommendation systems aim to bridge this gap by leveraging machine learning models to match research papers with the most relevant journals based on their content and scope. Recent advancements in natural language processing (NLP), particularly transformer-based models, have significantly improved the performance of such systems by providing robust semantic representations of textual data. Models like BERT (Bidirectional Encoder Representations from Transformers) [1] and its variants have demonstrated state-of-the-art results in various NLP tasks, including text classification, semantic similarity, and recommendation systems.

This research investigates the role of advanced pre-trained language models and fine-tuning strategies in enhancing journal recommendation systems for distinct academic domains. The study focuses on three key objectives:

1. evaluating the performance of advanced models, such as all-MiniLM-L6-v2 [2] and BAAI/bge-small-en-v1.5 [3], against the baseline model, Distil-RoBERTa [4];
2. analyzing the impact of domain-specific fine-tuning on classification performance within and across domains, specifically using Computer Science and Biology datasets; and
3. comparing the effectiveness of fine-tuned models to directly leveraging pre-trained embeddings for classification tasks on isolated datasets.

The methodology builds upon prior approaches [5], where Distil-RoBERTa was fine-tuned using a contrastive learning framework for paper submission recommendation tasks within the Computer Science domain. The enhanced approach in this study employs advanced transformer-based models, including all-MiniLM-L6-v2 and BAAI/bge-small-en-v1.5, and expands the scope to include Biology as a second domain. Two distinct strategies are implemented: (1) fine-tuning the advanced models using contrastive learning on the Computer Science dataset, followed by training classification models for Computer Science and Biology datasets separately; and (2) directly using pre-trained embeddings from the original models to train classification models for both domains. These methods are evaluated to determine whether domain-specific fine-tuning yields superior performance compared to generic embeddings.

By integrating cutting-edge transformer models and exploring domain-specific fine-tuning, this study aims to advance the effectiveness and adaptability of journal recommendation systems. The research seeks to provide insights into the applicability of advanced NLP models for domain-specific tasks. Ultimately, the findings are expected to contribute to the development of more efficient and accurate frameworks for academic journal recommendations.

Related Work

Recommendation systems have gained significant importance in various domains, including academics, to assist researchers in identifying suitable venues for submitting their work. Early efforts in paper submission recommendation, such as the study by Wang et al. in 2018, utilized statistical methods like Chi-square and TF-IDF (Term Frequency-Inverse Document Frequency) for feature extraction and applied Softmax Regression for classification, achieving a Top-3 accuracy of 61.37% [6]. Subsequent approaches in 2020 introduced deep learning techniques with additional input features to improve performance. For example, Son et al. employed Logistic Linear Regression (LLR) and Multi-Layer Perceptrons on diverse feature combinations such as “titles,” “abstracts,” and “keywords,” reporting a Top-3 accuracy of 89.07% with the LLR model [7]. Advances in embedding methods, including GloVe [8] and FastText [9], led to more sophisticated models. Nguyen et al., in 2021, combined these embeddings with Convolutional Neural Networks and Long Short-Term Memory to propose the S2CFT model [10], which achieved a Top-3 accuracy of 90.21% and a Top-10 accuracy of 99.17% when integrating “titles,” “abstracts,” and “keywords.” That same year, the incorporation of external features like the “Aims and Scopes” of journals by Son et al. [11] acquired a Top-3 accuracy of 78.89% and a Top-

10 accuracy of 93.23%, demonstrating the potential of leveraging supplementary data to enhance recommendation accuracy.

In recent years, transformer-based architectures have revolutionized NLP tasks. Models like BERT [1] have been fine-tuned for document and sentence embeddings, improving their effectiveness in applications such as semantic textual similarity and classification. However, traditional fine-tuning approaches often face limitations in capturing fine-grained semantic relationships. Contrastive learning has emerged as a robust alternative, optimizing embeddings by pulling semantically similar samples closer and pushing dissimilar ones apart in the embedding space. Building on these advancements, the SimCPSR model [5] leverages contrastive learning to fine-tune a pre-trained Distil-RoBERTa transformer model [4]. By integrating document information (“titles,” “abstracts,” and “keywords”) with journals’ “aims and scopes,” SimCPSR achieves state-of-the-art results. The model surpasses previous methods, with a Top-3 accuracy of 81.12% and a Top-10 accuracy of 94.96%, highlighting its effectiveness in capturing semantic relationships and recommending relevant journals.

Despite the notable advancements in paper submission recommendation systems, several challenges remain unaddressed. Existing methods often rely on specific pre-trained language models (LMs) that, while effective, may not fully utilize the capabilities of more advanced transformer-based architectures. Furthermore, the systems primarily focus on single-domain datasets, specifically Computer Science, and the benefits of fine-tuning models on domain-specific datasets versus directly using pre-trained embeddings remain underexplored. As a result, this research aims to bridge these gaps by employing advanced LMs, extending the dataset to a new domain, namely Biology, and analyzing the effectiveness of fine-tuned models against their non-fine-tuned counterparts. Eventually, the study seeks to enhance the adaptability and accuracy of journal recommendation systems across diverse academic fields.

Methodology

This section details the methodology employed to enhance journal recommendation systems through advanced pre-trained language models and fine-tuning strategies. The study evaluates performance improvements by upgrading the baseline model, Distil-RoBERTa [4], with advanced models such as all-MiniLM-L6-v2 [2] and BAAI/bge-small-en-v1.5 [3]. The methodology further investigates the role of domain-specific fine-tuning and pre-trained embeddings in classification tasks across distinct academic domains, focusing on Computer Science and Biology datasets.

Baseline Approach

The original approach, SimCPSR model [5], utilizes the Computer Science dataset and employs a two-stage methodology for paper submission recommendation task using a pre-trained LM. The first stage involves fine-tuning the pre-trained LM with a contrastive learning approach, which allows the model to efficiently generate sentence embeddings for each document or sentence. Once fine-tuned, the model can be utilized for various downstream classification tasks, including recommending the most relevant journals for a given paper to increase the change of acceptance.

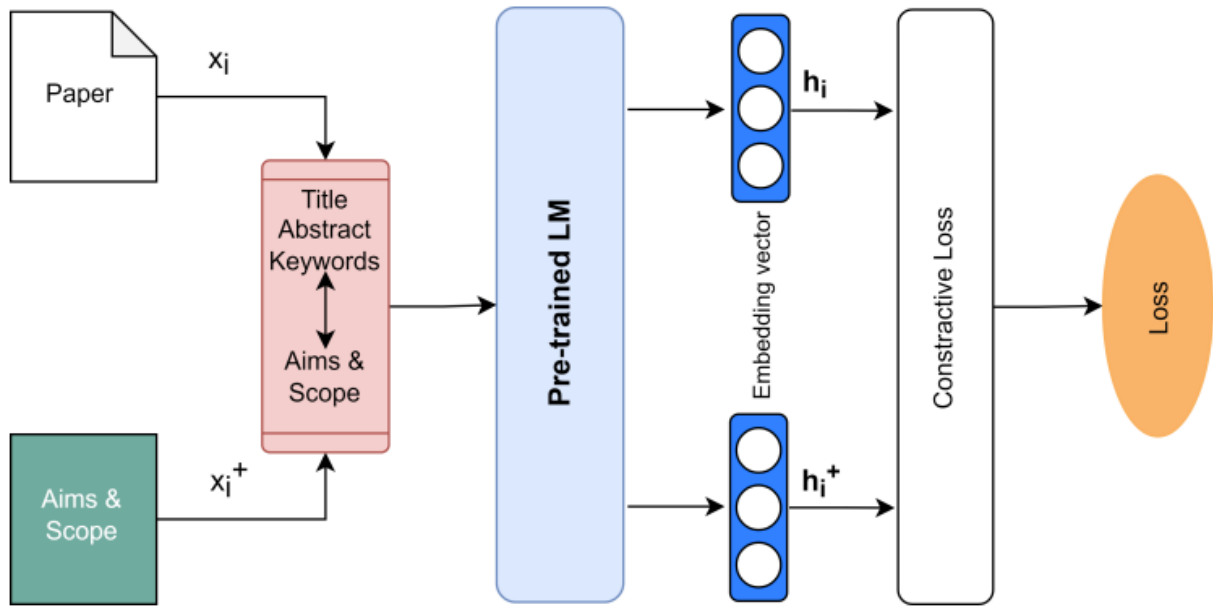


Figure 1. Model Architecture of the Fine-tuned Pre-trained Language Model [5]

In the fine-tuning stage, the model adopts a supervised contrastive learning framework to optimize semantic representations of papers and journals. Positive samples are constructed using journal attributes such as the journal's “*aims and scope*,” paired with a paper's “*title*,” “*abstract*,” and “*keywords*,” while negative pairs are created from unrelated journal attributes. Two semantically related samples, x_i (e.g., a paper's title, abstract, and keywords) and x_i^+ (e.g., the journal's aims and scope), are input into the pre-trained LM to generate their respective embedding vectors h_i and h_i^+ , as depicted in Figure 1. These embeddings are compared using a contrastive loss function, which minimizes the distance between related pairs while maximizing the separation between unrelated pairs. This fine-tuning process updates all parameters of the model, enhancing its ability to capture semantic similarity and contextual relevance, thereby improving performance in downstream classification tasks.

The fine-tuned Distil-RoBERTa then serves as the core encoder that processes various combinations of features, either derived from the paper's metadata or a mix of paper information and journal-specific data. The experiments explore two primary use cases:

1. **Paper Information-Based Models:** Key information is extracted from each paper, including its “*title*,” “*abstract*,” and “*keywords*,” which are concatenated into seven feature combinations: Title (T), Abstract (A), Keywords (K), Title + Abstract (TA), Title + Keywords (TK), Abstract + Keywords (AK), and Title + Abstract + Keywords (TAK). The model architecture follows the approach illustrated in Figure 2. The fine-tuned LM encodes these combinations into a high-dimensional embeddings, which are processed by a linear layer with ReLU activation and dropout to mitigate overfitting. The final softmax layer produces probabilities for each journal, and the Top K journals with the highest probability of acceptance are selected for the top recommendations.

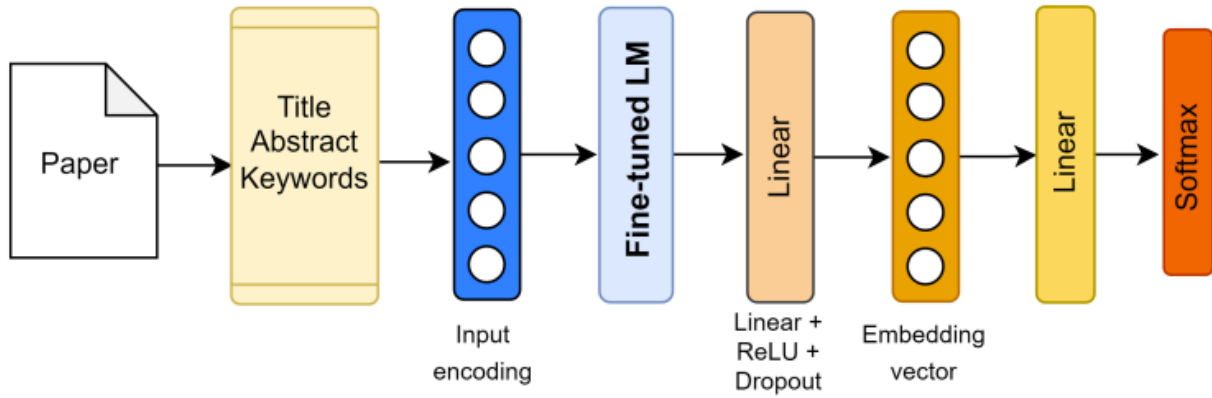


Figure 2. Classification Architecture for the Paper Information-Based Models [5]

2. **Paper and Journal Information-Based Models:** The feature set is extended by incorporating the journals' "*aims and scopes*" along with the paper's information, which leads to seven new feature combinations: Title + Aims and Scopes (TS), Abstract + Aims and Scopes (AS), Keywords + Aims and Scopes (KS), Title + Abstract + Aims and Scopes (TAS), Title + Keywords + Aims and Scopes (TKS), Abstract + Keywords + Aims and Scopes (AKS), and Title + Abstract + Keywords + Aims and Scopes (TAKS). The model architecture, as demonstrated in Figure 3, is composed of two parallel pathways feeding into a central processing branch. One pathway encodes external journal metadata (e.g., aims and scopes) into embeddings using a fine-tuned LM, followed by dimensionality reduction via a linear layer. The second pathway processes the paper's core elements (e.g., title, abstract, and keywords), mirroring the approach used in the previous model. The embeddings from both pathways undergo similarity evaluation through cosine similarity, and the resulting outputs are concatenated with the paper embedding to form a comprehensive feature vector. This vector is passed through a linear layer and a softmax classifier to generate journal predictions. The top-ranked journals are selected and recommended based on their predicted relevance.

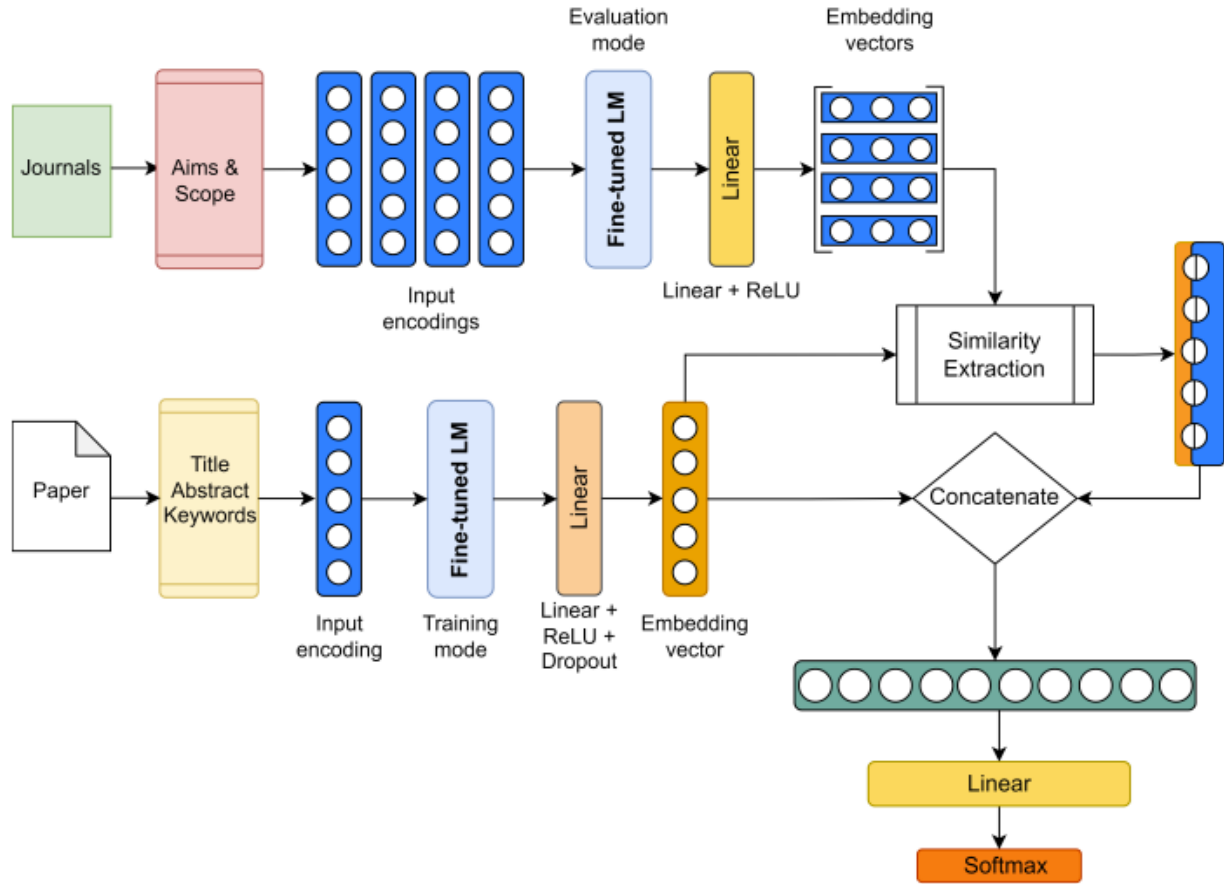


Figure 3. Classification Architecture for the Paper and Journal Information-Based Models [5]

This two-stage methodology combines both internal paper features and external journal characteristics, leveraging the strengths of the fine-tuned LM for improved classification accuracy and relevance in recommending journals.

Proposed Improvements

To boost recommendation accuracy and model performance, two key enhancements are proposed. First, we aim to upgrade the current pre-trained ML by replacing it with a more advanced transformer-based architecture. Candidate models include all-MiniLM-L6-v2 [2] and BAAI/bge-small-en-v1.5 [3]. These models are lightweight yet powerful, offering efficient computation while maintaining high-quality embeddings. all-MiniLM-L6-v2 demonstrates exceptional proficiency in capturing semantic similarity between sentences, making it particularly well-suited for applications such as clustering and semantic search [2]. On the other hand, BAAI/bge-small-en-v1.5 exhibits robust performance across a diverse array of tasks, including text classification and comprehension, thereby rendering it highly versatile for numerous applications [12]. Nonetheless, both models may encounter limitations in processing extended sequences and may also reflect biases inherent in their training datasets [13]. The final model selection will be based on performance benchmarks, computational feasibility, and compatibility with our system, with the goal of maximizing the quality of recommendations

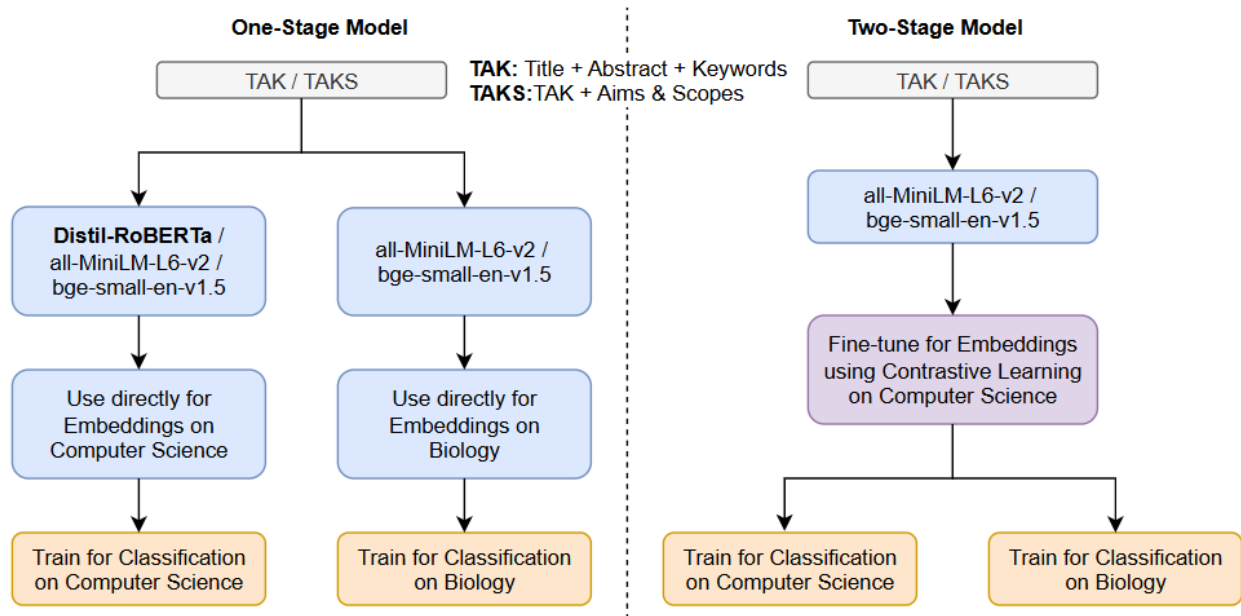


Figure 4: Architecture of the Proposed Improvements

Additionally, to analyze domain-specific performance and the effectiveness of fine-tuned models, datasets from Computer Science and Biology are utilized. This process, as presented in Figure 4, comprises two approaches: (1) the one-stage model, which utilizes the original pre-trained embeddings, without fine-tuning, to train classification models on the Computer Science and Biology datasets separately, and (2) the two-stage model, which fine-tunes advanced models on the Computer Science dataset and then uses them to train classification models for both datasets. Specifically, in the Computer Science domain, the models employed include Distil-RoBERTa-base, all-MiniLM-L6-v2, and bge-small-en-v1.5, with the latter two fine-tuned on the Computer Science dataset, as Distil-RoBERTa-base requires significant computational and memory resources, which are limited in this setting. In the Biology domain, only all-MiniLM-L6-v2 and bge-small-en-v1.5 models are used, both with and without fine-tuning on the Computer Science dataset, due to the same computational constraints. This combination of fine-tuned and non-fine-tuned embeddings across domains enables a thorough evaluation of the models' adaptability and effectiveness in distinct academic fields.

In terms of dataset construction, the new dataset for the Biology domain is collected through web scraping from Springer¹, alongside the Computer Science papers used in Son et al.'s work [5]. It also comprises paper details such as the "title," "abstract," and list of "keywords," as well as the corresponding journal's "aims and scopes." Data scraping is conducted by searching for relevant journals within specified domains, utilizing targeted keywords and crawling over thousands of papers, along with their associated aims and scopes. To ensure data quality, journals without aims and scopes are excluded, as well as papers lacking abstracts or keywords. As a result, the final dataset includes 583,614 papers from 724 journals, distributed as follows: 414,512 papers and 351 journals from Computer Science, and 169,102 papers and 193 journals from Biology. This

¹ <https://www.springer.com>

curated dataset serves as a comprehensive resource for assessing the domain-specific performance and effectiveness of fine-tuned models in classification tasks within the Computer Science and Biology fields.

Implementation

To ensure the highest model performance, robust data preprocessing is essential. This stage follows the optimal procedures outlined in Son et al.'s work [5], focusing on cleaning and standardizing input data. The preprocessing pipeline involves common NLP steps: converting text to lowercase, removing punctuation, URLs, and stop words (using the Natural Language Toolkit), and eliminating unnecessary spaces and non-alphabetic characters. These steps are consistently applied to both paper metadata (e.g., title, abstract, and keywords) and journal aims and scopes, ensuring uniformity across datasets. Comprehensive preprocessing plays a vital role in enhancing model accuracy and adaptability across diverse academic fields.

For efficient model training and deployment, the experimental setup is implemented using PyTorch [14], with HuggingFace Transformers [15] facilitating seamless integration of pre-trained models. The development environment comprises hardware such as an NVIDIA GPU or MacBook GPU, equipped with a minimum of 32GB of RAM, to provide effective execution of complicated computational operations. The model training approach starts with fine-tuning an embedding model by contrastive loss, with a decay rate of 0.86 and a learning rate of $5e^{-6}$. The recommendation model training procedure encompasses 8 epochs, employing training and validation data loaders to repeatedly assess performance. The model generates predictions and evaluates loss through a specified loss function, while the optimizer adjusts the model's weights according to the calculated gradients. Accuracy is assessed at multiple Top-K levels, offering insights into the model's efficacy in obtaining pertinent recommendations. Validation losses are consistently checked to ensure the preservation of the optimal performing model. This setup establishes a robust and efficient foundation for training and fine-tuning, enabling precise evaluation of the model's effectiveness and scalability through the use of contemporary hardware and advanced learning methodologies.

To produce high-quality embeddings, the classification architecture leverages advanced tokenization and encoding processes. Input data is tokenized using the selected model's tokenizer, with sequence lengths standardized through padding or truncation. Based on the results reported in Son et al.'s work [5], the best two feature combinations of TAK and TAKS are identified as optimal and are encoded into fixed-size embeddings. These embeddings are passed through fully connected layers with ReLU activation to refine their representation, ultimately generating probabilities for each journal. A softmax layer is used to compute probabilities for recommending the Top K journals, aligning with the Accuracy@K evaluation metric described by Son et al. [5]. This architecture ensures the system effectively prioritizes the most relevant journals.

The effectiveness of the model is rigorously evaluated using Accuracy@K [5], a metric that measures the percentage of relevant journals appearing in the top K recommendations. K values are set to 1, 3, 5, and 10, providing a comprehensive understanding of the model's ranking capabilities. Equation 1 illustrates the calculation process:

$$Accuracy@K = \frac{\text{Number of relevant items in the top } K}{K} \quad (1)$$

This evaluation metric offers critical insights into the model’s capacity to generate accurate and relevant journal recommendations, supporting the overarching goal of optimizing the paper submission process.

Results and Evaluation

The performance of the journal recommendation system is evaluated using the Accuracy@K metric, focusing on Top-1, Top-3, Top-5, and Top-10 journal recommendations. Two advanced transformer-based models—all-MiniLM-L6-v2 and BAAI/bge-small-en-v1.5—are compared against the baseline Distil-RoBERTa. The evaluation is conducted separately on Computer Science and Biology datasets to assess the effects of domain-specific fine-tuning on classification performance both within and across domains. Additionally, the study examines the performance of fine-tuned models compared to using pre-trained embeddings directly for classification tasks in independent datasets.

1. Domain	Model	Feature	Top 1	Top 3	Top 5	Top 10
Computer Science (CS)	Distil-RoBERTa-base (epoch 3)	TAK	0.520	0.819	0.890	0.951
	Distil-RoBERTa-base (epoch 3)	TAKS	0.515	0.810	0.883	0.946
	all-MiniLM-L6-v2	TAK	0.517	0.810	0.885	0.948
	all-MiniLM-L6-v2	TAKS	0.507	0.797	0.874	0.941
	all-MiniLM-L6-v2 (finetune on CS)	TAK	0.509	0.803	0.880	0.947
	all-MiniLM-L6-v2 (finetune on CS)	TAKS	0.508	0.797	0.875	0.947
	bge-small-en-v1.5	TAK	0.530	0.825	0.897	0.955
	bge-small-en-v1.5	TAKS	0.533	0.826	0.894	0.954
	bge-small-en-v1.5 (finetune on CS)	TAK	0.524	0.815	0.890	0.953
	bge-small-en-v1.5 (finetune on CS)	TAKS	0.525	0.815	0.890	0.952
Biology	all-MiniLM-L6-v2	TAK	0.765	0.898	0.935	0.969
	all-MiniLM-L6-v2	TAKS	0.764	0.896	0.935	0.968
	all-MiniLM-L6-v2 (finetune on CS)	TAK	0.773	0.901	0.939	0.971
	all-MiniLM-L6-v2 (finetune on CS)	TAKS	0.767	0.897	0.934	0.966
	bge-small-en-v1.5	TAK	0.787	0.912	0.946	0.975
	bge-small-en-v1.5	TAKS	0.785	0.911	0.946	0.975
	bge-small-en-v1.5 (finetune on CS)	TAK	0.788	0.913	0.946	0.975
	bge-small-en-v1.5 (finetune on CS)	TAKS	0.783	0.909	0.944	0.975

Table 1: Evaluation Results

According to the evaluation results presented in Table 1, several key findings emerge:

- The advanced models, particularly BAAI/bge-small-en-v1.5, consistently outperformed the baseline Distil-RoBERTa model in the Computer Science domain. Additionally, the pre-trained embeddings from BAAI/bge-small-en-v1.5 demonstrates exceptional performance on the Biology dataset.

3. Contrary to Son et al.'s findings [5], incorporating the journal's "aims and scopes" feature into the model does not boost its performance. The evaluation results reveal that models incorporating the journal's aims and scopes consistently underperformed compared to those relying solely on the paper metadata.
4. The application of domain-specific fine-tuning on the Computer Science dataset does not yield consistent or significant improvements within the Computer Science domain while providing only minimal benefits when applied to the Biology dataset.
5. Directly using pre-trained embeddings from BAAI/bge-small-en-v1.5 proves to be a highly effective strategy, especially within the Computer Science domain. This is demonstrated by the competitive performance observed from the BAAI/bge-small-en-v1.5 model compared to the fine-tuned version.

The integration of advanced pre-trained language models significantly improved journal recommendation performance in both the Computer Science and Biology domains. In Computer Science, BAAI/bge-small-en-v1.5 achieves a Top-1 accuracy of 0.530 with TAK and 0.533 with TAKS, surpassing both Distil-RoBERTa and all-MiniLM-L6-v2. This model consistently outperforms others across Top-K metrics, regardless of whether fine-tuning was applied. On the other hand, the models applied to Biology domain demonstrate even stronger performance across all models, metrics, and features. all-MiniLM-L6-v2 achieves a Top-1 accuracy of 0.765 with TAK and 0.764 with TAKS, while fine-tuning on the Computer Science dataset slightly improved the performance. bge-small-en-v1.5 shows superior accuracy, with a Top-1 accuracy of 0.787 for TAK and 0.785 for TAKS, though fine-tuning on the Computer Science dataset does not significantly change the results. Overall, BAAI/bge-small-en-v1.5 consistently outperforms both baseline models and all-MiniLM-L6-v2 in both the TAK and TAKS configurations, proving its robustness and adaptability across different domains.

Additionally, despite expectations, the inclusion of the journal's "*aims and scopes*" feature does not lead to improved model performance. This is evident from the results, where models using TAKS repeatedly fell short when measured against those using TAK. For example, Distil-RoBERTa achieved a Top-1 accuracy of 0.520 on the Computer Science dataset with TAK, but only 0.515 with TAKS. Similarly, BAAI/bge-small-en-v1.5 shows a performance drop of approximately 0.003-0.005 across metrics when using TAKS instead of TAK. This finding suggests limitations in the use of cosine similarity to capture meaningful relationships between papers and journals. It is also possible that the paper metadata alone already provides sufficient contextual information for the model, and adding "aims and scopes" does not offer a substantial advantage in this case. Thus, the cosine similarity between the paper information and the journal's aims and scopes does not significantly enhance model performance, highlighting the importance of carefully select features and thoughtfully represent them to maximize their contribution to the model's effectiveness.

Besides, the results of domain-specific fine-tuning on the Computer Science dataset are less impactful than initially expected. Fine-tuning both all-MiniLM-L6-v2 and BAAI/bge-small-en-v1.5 using contrastive learning on the Computer Science dataset does not result in significant performance gains on the Computer Science test set. In some cases, the performance even slightly decreases after fine-tuning. Furthermore, the cross-domain transferability of the fine-tuned models is limited. When applied to the Biology dataset, the fine-tuned models on Computer Science

domain exhibits lower performance compared to models utilizing their original pre-trained embeddings, particularly the BAAI/bge-small-en-v1.5 model. This suggests that the domain-specific knowledge gained through fine-tuning on the Computer Science dataset does not generalize well to the distinct features of the Biology domain. Variations in vocabulary, writing styles, and research focuses between the two domains likely contributed to this outcome, emphasizing the challenges of transferring fine-tuned knowledge across domains.

As a result, among the evaluated models, BAAI/bge-small-en-v1.5 (TAK) is the best-performing model, demonstrating superior accuracy and generalizability across both domains. Its ability to achieve high Top-K accuracy, coupled with consistent performance across datasets, underscores its robustness and practical applicability. These findings validate the potential of advanced pre-trained transformers in optimizing journal recommendation systems. Besides, although fine-tuning and feature expansion offer incremental benefits, the broader success lies in leveraging high-quality embeddings and robust pre-trained models. Hence, future advancements in journal recommendation systems should prioritize the development of models that effectively balance generalization and domain-specific adaptability, leveraging high-quality embeddings while carefully optimizing feature selection and representation strategies.

Additionally, despite the promising outcomes, a key limitation of the proposed models lies in their adaptability across diverse academic fields, as they often over-fit to domain-specific features. Addressing this challenge requires future efforts to explore alternative methods for leveraging journal-specific metadata and refining fine-tuning strategies to enhance generalization capabilities. One promising avenue is the use of domain adaptation techniques, such as adversarial training with gradient reversal, to learn domain-invariant features. This approach integrates a domain classifier alongside the main task, where the classifier predicts the data's domain, and the gradient reversal layer encourages feature representations to be less domain-specific. By focusing on shared patterns across domains, the model can improve its generalization across different fields. Additionally, fine-tuning on labeled datasets from target domains can further refine performance by adapting the model to the unique characteristics of specific domains. Combining adversarial training to foster generalization with targeted fine-tuning to enhance specialization provides a balanced and effective strategy for improving cross-domain adaptability, ensuring the model remains robust and effective across diverse academic disciplines.

Conclusion

This study demonstrates the potential of advanced transformer-based models in optimizing journal recommendation systems across various domains. The analysis revealed that pre-trained embeddings from models like BAAI/bge-small-en-v1.5 significantly outperform other transformer models, including fine-tuned counterparts in some cases, highlighting their robustness and adaptability. Surprisingly, incorporating journal-specific metadata, such as “*aims and scopes*,” does not enhance performance, suggesting the sufficiency of core paper features like “*titles*,” “*abstracts*,” and “*keywords*” for recommendation tasks and illustrating the ineffective capturing of semantic relationships between the paper and journal information. Additionally, fine-tuning on the Computer Science dataset fails to improve performance within the same domain and when applied across domains (e.g., Biology), fine-tuned models show only minimal gains compared to

directly leveraging pre-trained embeddings. These findings indicate that domain-specific fine-tuning may not be highly effective, raising concerns about the computational cost when simpler methods like pre-trained embeddings perform similarly or even better.

Besides, the results also emphasize the importance of feature selection and alternative strategies to improve generalizability. Techniques such as adversarial training could help develop domain-invariant features, thereby enhancing cross-domain adaptability. Furthermore, approaches like meta-learning or targeted fine-tuning on smaller, diverse datasets may offer a balanced solution for achieving both generalization and specialization. Overall, this work advances the field of academic recommendation systems by providing valuable insights into the application of cutting-edge NLP models for diverse and dynamic research needs. Future efforts should prioritize the development of adaptive methodologies, diverse datasets, and refining fine-tuning strategies to optimize both domain-specific and cross-domain performance.

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